How Long Is a Presence?
Refuting a Geometric Derivation of the Alert Collapse Window in Multi-Analytics CCTV

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Abstract

A CCTV platform running several analytics over the same scene emits one alert per detection unless it collapses them. The collapse predicate — same entity, same window — is standard: it is Debar and Wespi [9]'s aggregation predicate under a normalised record schema of the IDMEF [10] and OCSF [22] lineage, evaluated over keyed session windows in the sense of Akidau et al. [3]. The predicate is not our contribution and we do not claim it. The open parameter is the window W, and our deployment set it to 120 s with a derivation: camera field-of-view depth divided by permitted speed.

That derivation is wrong, and our own data refutes it. Computed per camera across a 200-camera estate, the geometric dwell time is 1.27 to 9.63 s with a median of 3.66 s — one and a half orders of magnitude below the deployed constant. The deployed 120 s over-collapses relative to its own stated justification at every camera in the estate (200 of 200). Worse, the geometric quantity is not the right one: what the window must span is the interval between successive *observations* of an entity, which includes revisits and re-detections, not the time to traverse one field of view.

Measured on the operating curve, the knee sits at 15–30 s. Moving from 15 s to 120 s buys a further 2.8% reduction in alerts and costs 337 additional masked incidents, taking distinct-incident recall from 0.9961 to 0.9680. We give the corrected procedure: choose W from the inter-observation-time distribution by a cost-weighted operating point, with Satopää et al. [23] as the parameter-free default. Separately, we confirm that an entity-keyed predicate spanning modalities beats per-modality deduplication by 10.5% to 43.5% as the dual-tagged incident share rises from 0.15 to 1.0, at no recall cost, and that a confidence-voting alternative achieves the largest raw reduction (88.1%) only by dropping 63.5% of incidents and is not a usable operating point.

All results are SYNTHETIC, from a seeded generator over 59,984 detections and 12,000 incidents. §8 states what real operator-marked ground truth must show.

1 Introduction

Run automatic number-plate recognition, face matching and region-occupancy detection over the same camera and the same vehicle generates three detections a second. Emit one alert each and the control room receives five alerts per incident; the literature on alert fatigue [26, 4, 25] and the older process-control work on alarm floods [28, 17, 12] is unanimous that this is the failure mode that ends the deployment.

The fix is to collapse: emit one alert per entity per window. The question that decides whether the fix works is how long the window is. Too short and the operator gets duplicates; too long and a genuinely new incident involving the same entity is masked by a still-open alert. This is a two-sided error and it has an operating curve.

Our deployment chose $W = 120\,\mathrm{s}$ and documented a derivation for it: a camera's field of view is a few tens of metres deep, permitted speed is a few tens of km/h, so a vehicle is present for on the order of a minute or two. This paper reports that the derivation does not survive contact with the estate's own parameters, explains why the geometric quantity is the wrong one in principle as well as in magnitude, and replaces it with a procedure.

1.1 What is and is not claimed

Not claimed: the normalised alert record (IDMEF [10], OCSF [22], JDL level-2 common referencing [24, 16]); the same-entity-same-window aggregation predicate [9, 18, 27]; keyed session windowing [3, 8]; or duplicate detection as a discipline [14, 11]. All of this is background and §2 says so in its first sentence.

Claimed: (i) the geometric derivation of W is refuted by the estate's own geometry, quantitatively (§4); (ii) the measured knee is 15–30 s and the deployed value sits far past it, with the cost quantified (§6); (iii) a defensible replacement procedure for choosing W (§5); (iv) the cross-modal entity-keyed predicate's marginal value over per-modality deduplication, measured against the dual-tagging rate (§6.1).

2 Background, all of it established

Nothing in this section is a contribution. Alert normalisation into a common record with entity, time, source and confidence fields is IDMEF [10] and, in current practice, OCSF [22]; the fusion framing is JDL level 2 [24], surveyed by Khaleghi et al. [19]. Aggregating alerts that share a key within a time window is Debar and Wespi [9], refined by Julisch [18] and Ning et al. [21], with a probabilistic variant in Valdes and Skinner [27]. Evaluating such a predicate over unbounded streams with per-key session windows is Akidau et al. [3]. Session-gap estimation from inter-event times has its own literature in web analytics [6, 15, 20], and in process control the same problem is the alarm dead-band and delay-timer design question [1, 2, 29].

We adopt all of it. The gap is that none of these sources tells a CCTV operator what number to type in, and the one derivation our deployment invented for itself is wrong.

3 The collapse predicate as deployed

A detection is a tuple (e, m, c, t, s): entity key, modality, camera, timestamp, score. The predicate is entity-keyed and modality-agnostic:

> Emit an alert for detection d iff no alert with the same entity key e is open; an alert opened at t_0 is open over $[t_0, t_0 + W)$.

Window semantics are *anchored*, not merging: a detection inside an open window does not extend it. This matters and is under-reported in the literature. Merging semantics turn a stream of detections at a busy junction into one unbounded alert; anchored semantics bound the suppression at W by construction. Every number in this paper is under anchored semantics.

Two error modes follow. A *redundant alert* is a second alert for an incident already alerted. A *masked incident* is a distinct incident that produced no alert because an unrelated earlier incident with the same entity key held the window open. We report distinct-incident recall for the second, because plain incident recall is 1.0 everywhere and hides the failure.

4 Why the geometric derivation fails

Table 1 is the refutation. Field-of-view depth divided by permitted speed gives seconds, not minutes, at every camera in the estate. If the deployment had applied its own stated rule it

Quantity	**Value**
Geometric dwell, minimum over estate	1.27 s
Geometric dwell, median over estate	3.66 s
Geometric dwell, maximum over estate	9.63 s
Deployed constant W	120 s
Cameras where the constant exceeds the geometric value	200 of 200

Table 1: **Geometric window per camera against the deployed constant (synthetic, 200 cameras).** W_{geom} = FOV depth/permitted speed. The deployed constant exceeds its own stated derivation by a factor of 12 at the most generous camera and 94 at the tightest.

would have set W between 1 and 10 s, and per camera rather than globally. It did not; it set 120 s. The derivation was therefore not the reason for the value, whatever the design document says.

The magnitude error is the smaller problem. The conceptual error is that dwell time is not the quantity the window must span. The window exists to suppress *repeat observations of the same entity that belong to the same operational event.* Those arise from (a) multiple analytics firing on one pass, (b) repeated detections across frames during one pass, and (c) the entity leaving and returning — circling a block, stopping and restarting, being observed by an adjacent camera whose alerts share the entity key. Case (c) dominates the tail and has nothing to do with how long one field of view is. A derivation from geometry cannot see it. The correct input is the empirical inter-observation-time distribution.

5 Choosing W properly

Four families of method apply, and the choice among them is a value judgement that should be made explicitly rather than buried in a constant.

Cost-weighted operating point. Treat window selection as threshold selection on a two-sided error curve [13]. Assign a cost λ_r to a redundant alert (operator seconds) and λ_m to a masked incident (a missed event), and minimise $\lambda_r R(W) + \lambda_m M(W)$. This is the framing we lead with, because it makes the deployment's values legible and lets another deployment substitute its own. Axelsson [5] is the reason the two costs are not comparable by intuition.

Mixture crossover. Fit a two-component mixture to inter-observation times — within-event and between-event — and set W at the posterior crossover. This is the web-analytics session-gap method [6, 20] and is the most principled where the two populations are separable.

Knee detection. Apply Satopää et al. [23] to the reduction-versus-masking curve. Parameter-free, and the right default where no cost model exists.

Spec-based timer design. The process-control approach [1, 2, 29]: set the dead-band from a specified maximum acceptable detection delay. Appropriate where a regulator states the delay bound.

All four are defensible. Dividing a distance by a speed is not one of them.

6 Evaluation

Table 2 is the paper's central result. Read it in two parts.

Below the knee, collapse is nearly free. Going from 0 to 15 s removes 47,976 of the 47,984 redundant alerts and masks 47 incidents: 80.1% fewer alerts for 0.4% of distinct-incident recall. No deployment should decline this.

Above the knee, the curve is almost flat in the benefit and steep in the cost. From 15 s to 120 s, alerts fall by a further 335 (2.8% of the remainder, 0.5 percentage points of

W (s)	Alerts	Reduction vs. W=0	Masked incidents	Distinct-incident recall
0	59,984	—	0	1.0000
5	12,962	78.4%	19	0.9984
15	11,961	80.1%	47	0.9961
30	11,915	80.1%	87	0.9928
60	11,844	80.3%	164	0.9863
120	11,626	80.6%	384	0.9680
180	11,345	81.1%	672	0.9440
240	11,105	81.5%	904	0.9247
480	10,378	82.7%	1,630	0.8642
900	9,288	84.5%	2,716	0.7737
1800	9,012	85.0%	2,988	0.7510

Table 2: **The operating curve (synthetic, 59,984 detections, 12,000 incidents).** Redundant alerts are already down to 8 at W=15 s from 47,984 at W=0. Everything bought beyond 15 s is bought from recall.

Method	Alerts	Reduction	Incident recall	Distinct recall	Alerts/incident
naive (one per detection)	59,984	—	1.000	1.000	5.00
per-modality dedup	15,788	73.7%	1.000	0.9635	1.32
confidence voting	7,130	88.1%	**0.365**	0.3483	1.63
entity-agnostic collapse	11,626	80.6%	1.000	0.9680	0.97

Table 3: **Collapse strategies (synthetic), W = 120 s.** Confidence voting achieves the largest reduction by discarding 63.5% of incidents entirely. It is reported because it is the obvious alternative, and rejected because no control room can run at 0.365 incident recall.

total reduction) while masked incidents rise from 47 to 384, an eightfold increase. Recall drops from 0.9961 to 0.9680. The deployed value therefore sits well past the knee: it pays 337 masked incidents for a reduction an operator would not notice. At 1,800 s the trade is grotesque — 85.0% reduction against 0.7510 recall — but the shape of the curve is already established by 120 s.

Satopää et al. [23] applied to the (reduction, masking) curve returns a knee in 15–30 s, which is where we recommend the deployment move, subject to the cost weighting in §5.

6.1 Cross-modal collapse against per-modality deduplication

Table 4 isolates the one design decision in the predicate that is genuinely ours to defend: keying on the entity rather than on (entity, modality). The benefit is exactly proportional to how often an incident is seen by more than one analytic, from zero at 0% dual tagging to 43.5% at 100%. Our estate runs at roughly 40% dual tagging (4,859 of 12,000 incidents), placing the real benefit near 24%. A deployment running one analytic per camera should not adopt this and gains nothing from it; that conditional is the honest form of the claim.

7 Related work

Covered in §2 and conceded there. The one comparison we owe is Valdes and Skinner [27], whose weighted-similarity aggregation is the principal alternative to a hard entity key: it fuses on graded feature agreement rather than exact match. It would handle entity-key errors — a misread plate — which our predicate cannot, and a deployment with noisy keys should prefer it. We do not claim superiority over it; our estate's keys come from ANPR with a confidence gate, and exact matching is adequate there. Cheng et al. [7] and Ye et al. [30] are the alternatives

Dual-tagged share	Per-modality alerts	Entity-agnostic alerts	Extra reduction
0.00	11,654	11,654	0.0%
0.15	12,990	11,633	10.5%
0.30	14,415	11,671	19.0%
0.45	15,715	11,605	26.2%
0.70	17,973	11,637	35.3%
1.00	20,614	11,644	43.5%

Table 4: **Marginal value of crossing modalities (synthetic).** Recall is 1.000 for both methods at every share. The entity-agnostic column is flat because it is insensitive to how many modalities tagged the incident; that insensitivity is the entire benefit.

for graded matching across cameras.

8 Limitations

1. **Synthetic ground truth.** "Distinct incident" is a label our generator emits. In an estate, whether two appearances of one vehicle forty seconds apart are one incident or two is an operator judgement, and the whole curve in Table 2 depends on that judgement. The replication that matters is operator-marked incident boundaries on real footage; if operators mark a longer typical event than our generator, the knee moves right and 120 s may be defensible on grounds we did not test. The geometric refutation in §4 does not depend on this, since it compares the deployment against its own stated rule.

2. **Anchored semantics.** Under merging semantics all of these numbers change and the masking cost grows without bound.

3. **One estate, one entity-key source.** Key errors are not modelled.

4. **A single global W.** Table 1 shows the geometric quantity varies 7.6× across cameras. Whether a per-camera W pays is not answered here.

5. **Case grouping is a different problem.** An operator may want all alerts for one entity grouped in the interface for hours while still being notified of each. That is a presentation decision and it is arguably the right answer to the fatigue problem; this paper addresses suppression, not grouping.

9 Conclusion

The collapse predicate is not new and we do not claim it. Its window is the parameter that decides whether it helps, and the derivation our deployment used for that window — field-of-view depth over permitted speed — is refuted by the estate's own camera parameters at all 200 cameras, by a factor of 12 to 94, and is measuring the wrong quantity besides. On the operating curve the knee is at 15–30 s; the deployed 120 s pays 337 additional masked incidents for a 2.8% further reduction in alerts. We recommend deployments derive W from the inter-observation-time distribution under an explicit cost weighting, publish the weighting, and stop deriving it from geometry.

Reproduction

```
cd code
pip install -r requirements.txt
```

```
python -m pytest -q
python -m prresearch.p5_fusion.experiment
```

Fixed seeds; `results/p5_fusion.json` is byte-identical across runs. Code and data: `https://github.com/amitduabits/PRAHARI-P5-EventSchema`

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